

Synopsis

Database and Management Systems

Group project

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Project title:

**Renewable Energy Trading and Carbon Credit
Marketplace System**

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Introduction

The increasing concern over climate change has led to the development of carbon credit systems worldwide. Carbon credits are tradeable certificates issued to organizations that reduce carbon emissions by generating renewable energy. These credits can be sold to companies that exceed their emission limits.

The Renewable Energy Trading and Carbon Credit Marketplace System is designed to manage renewable energy production records, carbon credit issuance, trading transactions, and compliance monitoring using a structured database system.

The motivation for selecting this problem is to model a real-world sustainability and financial system that requires strong data integrity, transactional control, and regulatory verification.

Using a Database Management System (DBMS) instead of a file-based system provides:

- Data consistency and integrity
- Reduced redundancy
- Secure transaction management
- Multi-user access control
- Efficient querying and reporting
- Enforcement of constraints

A file-based system would not efficiently handle large transactional data, credit tracking, and regulatory verification processes.

Problem Statement

Many organizations generate renewable energy but require a structured mechanism to:

- Record energy production
- Issue carbon credits accurately
- Trade credits transparently
- Prevent duplicate credit issuance
- Track compliance with regulatory requirements

Manual or loosely structured systems may lead to:

- Data redundancy
- Credit miscalculation
- Fraudulent transactions
- Lack of audit trails
- Inconsistent compliance reports

The scope of database usage in this system includes:

- Managing company and plant records
- Recording and verifying energy production
- Issuing and tracking carbon credits
- Managing credit trading transactions
- Generating compliance reports

Objectives of the Project

The main objectives of this project are:

1. To design a database using Entity–Relationship (ER) data modeling.
2. To convert the ER model into relational tables.
3. To apply normalization up to 3NF.
4. To implement the database using SQL (DDL, DML, Queries).
5. To implement PL/SQL constructs such as procedures, functions, triggers, cursors.
6. To ensure data consistency using constraints and transaction management.

Scope of the Project

- **Functional Boundaries**

The system will include:

- Company registration and management
- Renewable energy plant management
- Energy production logging
- Carbon credit issuance
- Credit marketplace trading
- Compliance reporting

- **Types of Users**

- Company Admin**

- Registers plants
 - Uploads energy logs
 - Lists and buys credits

Regulatory Authority

- Verifies plant registrations
- Verifies energy production logs
- Audits compliance reports

Marketplace Admin

- Monitors transactions
- Handles disputes

• Backend Focus

The emphasis will be on:

- Database design
- SQL queries
- PL/SQL implementation
- Triggers and procedures
- Transaction control

Proposed System Flow

The proposed system will function as follows:

1. Companies register and submit plant details.
2. Regulatory authority approves the plant.
3. Companies submit energy production logs.
4. After verification, the system automatically calculates and issues carbon credits.
5. Companies list credits for sale in the marketplace.
6. Buyers purchase credits through secure transactions.
7. Wallet balances update automatically.
8. Compliance reports are generated.

Key Features

- Automated carbon credit issuance
- Secure credit wallet management
- Marketplace listing and trading
- Compliance report generation
- Role-based verification system
- Transaction consistency using ACID properties

The system improves:

- Efficiency (automation of credit issuance)
- Data integrity (constraints and triggers)
- Transparency (transaction records)
- Reliability (error handling and rollback mechanisms)

Database Design

• Entity-Relationship Diagram

Entities:

- Company
- EnergyPlant
- EnergyProductionLog
- CarbonCredit
- CreditWallet
- MarketListing
- Transaction
- RegulatoryAuthority
- ComplianceReport

Relationships

- One Company owns multiple EnergyPlants (1:N)
- One EnergyPlant generates multiple ProductionLogs (1:N)
- One verified ProductionLog generates one CarbonCredit batch (1:1)
- One Listing may have multiple Transactions (1:N)
- RegulatoryAuthority verifies logs and plants

Attributes include:

- Company: company_id (PK), name, industry, registration_no
- EnergyPlant: plant_id (PK), company_id (FK), type, capacity
- ProductionLog: log_id (PK), plant_id (FK), energy_kwh
- CarbonCredit: credit_id (PK), company_id (FK), credits_issued
- Transaction: txn_id (PK), buyer_id, seller_id, quantity

• Relational Schema

COMPANY(company_id PK, name, industry, registration_no UNIQUE)

ENERGY_PLANT(plant_id PK, company_id FK, type, capacity, status)

PRODUCTION_LOG(log_id PK, plant_id FK, energy_kwh, verification_status)

CARBON_CREDIT(credit_id PK, company_id FK, log_id FK, credits_issued)

MARKET_LISTING(listing_id PK, seller_id FK, credit_id FK, quantity, price)

TRANSACTION(txn_id PK, listing_id FK, buyer_id FK, quantity, total_amount)

Primary and foreign key constraints will ensure referential integrity.

Normalization

Functional Dependencies:

company_id → company_name, industry

plant_id → company_id, capacity

log_id → plant_id, energy_kwh
credit_id → log_id, credits_issued

Normalization Process:

- 1NF: Atomic attributes (no repeating groups)
- 2NF: Removal of partial dependency
- 3NF: Removal of transitive dependency

Database Implementation

- **SQL Implementation**

DDL Commands:

- CREATE TABLE
- ALTER TABLE
- DROP TABLE

DML Commands:

- INSERT
- UPDATE
- DELETE

SELECT Queries will include:

- Joins (Company + Plant + Credit)
- Subqueries (Top buyers)
- Aggregate Functions (SUM, AVG, COUNT)
- GROUP BY and HAVING
- Views (Active Listings View, Compliance Summary View)

PL/SQL Components

The project will implement:

- Stored Procedures:
 - Verify production logs
 - Generate carbon credits
 - Execute credit purchase
 - Generate compliance report
- Functions:
 - Calculate carbon credits based on energy produced
- Triggers:
 - Auto-issue credits after verification
 - Update wallet after transaction
 - Prevent duplicate credit issuance
- Cursors:
 - Generate periodic compliance summary
- Exception Handling:
 - Insufficient credit balance
 - Duplicate record insertion
 - Transaction failure

Transaction Management & Concurrency

The system will use:

- COMMIT
- ROLLBACK

- SAVEPOINT

To ensure:

- Atomicity (complete trade or none)
- Consistency (valid state before and after transaction)
- Isolation (multi-user trade safety)
- Durability (permanent storage of transactions)

Basic concurrency control mechanisms such as row-level locking will be utilized.

Tools & Technologies Used

- DBMS Software: PostgreSQL
- SQL & PL/SQL
- Frontend: NextJS
- Backend: Express (NodeJS)

Expected Outcomes

- Successful implementation of a normalized relational database.
- Efficient data retrieval using complex SQL queries.
- Automation using triggers and procedures.
- Improved data consistency and integrity.
- Demonstration of advanced DBMS concepts.